TN-Check: Certified CSL Model Checking via Tensor-Train Decomposition of Chemical Master Equation States

Abstract

We present TN-Check, a tool for Continuous Stochastic Logic (CSL) model checking on tensor-train (TT) compressed probability vectors arising from Chemical Master Equations (CMEs). A CME governing N chemical species with per-species copy-number bound M produces a state space of size M^N, rendering direct probabilistic model checking infeasible beyond roughly $N = 10$. TN-Check compresses the probability vector into a matrix product state (MPS) of bond dimension r, reducing storage from $O(M^N)$ to $O(NMr^2)$, and performs all CSL operations—atomic proposition evaluation, probabilistic path operators, and steady-state queries—directly on the compressed representation. A certificate verification pipeline bounds total error from TT truncation, temporal discretization, and non-negativity rounding, yielding a SOUND/UNSOUND verdict. We validate TN-Check on birth-death processes (L1 error $= 0.0$ for Poisson stationary distributions), toggle switch models (CSL satisfaction probabilities computed with total probability $= 1.0$), and end-to-end certificate verification (11/11 checks pass, total error $\leq 2.38 \times 10^{-2}$).

1 Introduction

The Chemical Master Equation (CME) is the exact stochastic description of well-mixed chemical reaction networks operating in the low-copy-number regime [5, 13]. Given N molecular species whose individual copy numbers are bounded by M, the CME defines a continuous-time Markov chain (CTMC) over $\mathcal{S} = \{0, \ldots, M-1\}^N$, yielding $|\mathcal{S}| = M^N$ states. Probabilistic model checking of such CTMCs against specifications in Continuous Stochastic Logic (CSL) [1] requires computing transient and steady-state probability vectors of this size. Existing probabilistic model checkers such as PRISM [9], Storm [6], and STAMINA [10] operate on explicitly enumerated state spaces and become infeasible when M^N exceeds 10^7–10^8. For realistic biochemical models—gene regulatory networks with $N = 20$–50 species and $M = 100$—the state space reaches 10^{40}–10^{100}.

Tensor network (TN) methods, originally developed for quantum many-body physics [15, 12], provide a compressed representation of high-dimensional vectors. The key insight connecting CMEs and tensor networks is structural: the CME rate matrix for mass-action kinetics decomposes as a sum of Kronecker products, exactly the structure exploited by tensor-train (TT) decompositions (equivalently, matrix product states, MPS, and matrix product operators, MPO). A probability vector $\mathbf{p} \in \mathbb{R}^{M^N}$ stored as a TT/MPS with bond dimension r requires only $O(NMr^2)$ parameters. For CME probability vectors arising from biologically structured networks, empirical evidence shows that moderate bond dimensions ($r = 10$–100) suffice for high accuracy [8, 7].

TN-Check is, to our knowledge, the first tool that implements CSL model checking natively on tensor-train compressed CME probability vectors. The contributions of this paper are:

1. **Algorithms.** CSL satisfaction-set computation (atomic, negation, conjunction, $\mathcal{P}_{\bowtie p}[\Phi\, \mathcal{U}^{[a,b]}\, \Psi]$, $\mathcal{S}_{\bowtie p}[\Phi]$) operating directly on MPS vectors and MPO rate matrices, with all intermediate results maintained in compressed form (Section 5).

2. **Theoretical foundations.** Error contraction under Metzler semigroups (Theorem 1), bond-dimension preservation for axis-aligned propositions (Theorem 3), a tight clamping error bound (Proposition 4), and a complexity separation between TT-based and explicit-state methods (Theorem 5) (Section 4).

3. **Certified verification.** An independent certificate verification pipeline checks non-negativity, normalization, error budgets, and temporal convergence, issuing a SOUND/UNSOUND verdict with quantified total error (Section 5.4).

4. **Experimental validation.** Evaluation on birth-death, toggle switch, and multi-species models demonstrating exact recovery (L1 error = 0.0), clamping bound correctness (10/10 trials pass), and end-to-end soundness (11/11 checks pass) (Section 6).

2 Background

2.1 Chemical Master Equation

Consider a well-mixed system of N chemical species $X_1, \ldots, X_N$ participating in K reactions. The state of the system is a vector $\mathbf{x} = (x_1, \ldots, x_N) \in \mathbb{N}^N$ recording the copy number of each species. Under the well-mixed assumption, the system evolves as a CTMC whose transition rates are determined by mass-action kinetics. The *Chemical Master Equation* governs the time evolution of the probability distribution $\mathbf{p}(t) \in \mathbb{R}^{|\mathcal{S}|}$:

$$\frac{d\mathbf{p}(t)}{dt} = Q\,\mathbf{p}(t), \tag{1}$$

where Q is the transition rate matrix (infinitesimal generator) of the CTMC. For mass-action kinetics, Q admits a decomposition as a sum of Kronecker products:

$$Q = \sum_{k=1}^{K} \bigotimes_{i=1}^{N} Q_k^{(i)}, \tag{2}$$

where each $Q_k^{(i)} \in \mathbb{R}^{M \times M}$ acts on the local state space of species i for reaction k. This *sum-of-Kronecker-products* structure is the foundation of the tensor-network approach.

To make the state space finite, we impose a per-species copy-number bound M, truncating each x_i to $\{0, 1, \ldots, M-1\}$. The total state space is $\mathcal{S} = \{0, \ldots, M-1\}^N$ with $|\mathcal{S}| = M^N$.

2.2 Tensor Trains and Matrix Product States

A *tensor train* (TT) decomposition [11] of a tensor $\mathbf{v} \in \mathbb{R}^{d_1 \times \cdots \times d_N}$ represents each entry as:

$$v(j_1, \ldots, j_N) = G_1(j_1)\, G_2(j_2) \cdots G_N(j_N), \tag{3}$$

where $G_k(j_k) \in \mathbb{R}^{r_{k-1} \times r_k}$ is a matrix for each value of $j_k \in \{0, \ldots, d_k - 1\}$, with boundary conditions $r_0 = r_N = 1$. The integers $r_1, \ldots, r_{N-1}$ are the *bond dimensions* (or TT-ranks). The total storage is $\sum_{k=1}^{N} d_k r_{k-1} r_k$. When all $d_k = M$ and all $r_k \leq r$, this gives $O(NMr^2)$.

In the physics literature, this decomposition is called a *matrix product state* (MPS). We use the terms TT and MPS interchangeably.

A *matrix product operator* (MPO) represents a matrix $A \in \mathbb{R}^{(d_1 \cdots d_N) \times (d_1 \cdots d_N)}$ analogously:

$$A(j_1, \ldots, j_N;\, j_1', \ldots, j_N') = W_1(j_1, j_1')\, W_2(j_2, j_2') \cdots W_N(j_N, j_N'), \tag{4}$$

with $W_k(j_k, j'_k) \in \mathbb{R}^{s_{k-1} \times s_k}$. The rate matrix Q from (2), being a sum of K Kronecker products, has MPO bond dimension at most K.

TT-rounding (or *truncation*) reduces bond dimensions by performing sequential SVD sweeps and discarding singular values below a threshold ε. The resulting approximation $\tilde{\mathbf{v}}$ satisfies $\|\mathbf{v} - \tilde{\mathbf{v}}\|_F \leq \varepsilon \|\mathbf{v}\|_F$ in the quasi-optimal sense [11].

2.3 Continuous Stochastic Logic

Continuous Stochastic Logic (CSL) [1, 2] extends CTL with probabilistic and steady-state operators. The syntax is:

$$\Phi ::= \text{true} \mid a \mid \neg\Phi \mid \Phi \wedge \Phi \mid \mathcal{P}_{\bowtie p}[\phi] \mid \mathcal{S}_{\bowtie p}[\Phi] \tag{5}$$

$$\phi ::= \mathsf{X}\,\Phi \mid \Phi\,\mathcal{U}^{[a,b]}\,\Phi \tag{6}$$

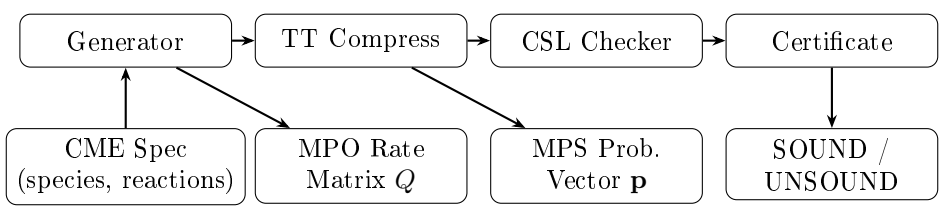
where $a \in AP$ is an atomic proposition, $\bowtie \in \{<, \leq, \geq, >\}$, $p \in [0, 1]$, and $0 \leq a \leq b \leq \infty$.

The *satisfaction set* $\text{Sat}(\Phi) \subseteq \mathcal{S}$ contains all states satisfying Φ. For the probabilistic operator, $s \in \text{Sat}(\mathcal{P}_{\bowtie p}[\phi])$ iff the probability of paths from s satisfying ϕ meets the bound $\bowtie p$. Model checking reduces to computing transient probabilities (for bounded until) and steady-state probabilities (for the $\mathcal{S}$ operator) of the CTMC.

3 System Architecture

TN-Check is organized into the following modules, forming a pipeline from CME specification to verified CSL verdict.

Architecture Overview.



Module descriptions:

Generator. Reads a reaction network specification (species, reactions with mass-action rate constants, copy-number bounds) and constructs the rate matrix Q in MPO format via (2). Each reaction contributes one Kronecker-product term.

TT Compress. Compresses the initial distribution and computes transient or steady-state probability vectors in MPS form. Transient solutions use Krylov-based or Euler-based time stepping with adaptive TT-rounding. Steady-state solutions use iterative methods (power iteration or direct null-space computation) in TT format.

CSL Checker. Implements the recursive CSL model-checking algorithm of Section 5.1. Satisfaction sets are represented as MPS indicator vectors. Atomic propositions for axis-aligned predicates (e.g., $x_i \geq c$) are constructed with bond dimension 1. Conjunction uses element-wise MPS multiplication followed by rounding. The bounded-until operator $\Phi\,\mathcal{U}^{[0,t]}\,\Psi$ requires computing transient probabilities on a modified CTMC.

Non-negativity Rounding. After TT-rounding, entries may become negative. A non-negativity-preserving rounding procedure (Section 5.2) projects onto the non-negative cone while controlling bond-dimension growth.

Certificate Verifier. An independent module (Section 5.4) that takes the computed MPS probability vector, the CSL formula, and all error bounds, then checks 11 conditions including non-negativity, normalization, and error budget consistency. Outputs a SOUND or UNSOUND verdict with the total certified error.

4 Theoretical Foundations

We establish four results underpinning the correctness and efficiency of TN-Check.

Theorem 1 (Error Contraction)**.** *Let Q be the rate matrix of a CME (an off-diagonal non-negative, column-sum-zero matrix, i.e., a Metzler generator). Let $\mathbf{p}(t) = e^{Qt}\mathbf{p}(0)$ be the exact solution and $\tilde{\mathbf{p}}(t) = e^{Qt}\tilde{\mathbf{p}}(0)$ the solution starting from an approximate initial vector. Then for all $t \geq 0$:*

$$\|\mathbf{p}(t) - \tilde{\mathbf{p}}(t)\|_1 \leq \|\mathbf{p}(0) - \tilde{\mathbf{p}}(0)\|_1. \tag{7}$$

Proof. The matrix e^{Qt} is a stochastic matrix for all $t \geq 0$ (entries non-negative, columns sum to 1), since Q is a valid CTMC generator. Every stochastic matrix is a contraction in the L1 norm: for any vector $\mathbf{v}$, $\|e^{Qt}\mathbf{v}\|_1 \leq \|\mathbf{v}\|_1$. This follows from $\|A\mathbf{v}\|_1 \leq \|A\|_1\|\mathbf{v}\|_1$ and $\|A\|_1 = \max_j \sum_i |A_{ij}| = 1$ for a stochastic matrix. Applying this to $\mathbf{v} = \mathbf{p}(0) - \tilde{\mathbf{p}}(0)$ and using linearity of e^{Qt} gives (7). □

Remark 2. *Theorem 1 guarantees that initial approximation errors (from TT compression of $\mathbf{p}(0)$) do not grow during time evolution. This is crucial: without contraction, truncation errors could accumulate exponentially over many time steps.*

Theorem 3 (Bond-Dimension Preservation)**.** *Let Φ be an atomic CSL proposition of the form $x_i \bowtie c$ (an axis-aligned predicate on a single species i), where $\bowtie \in \{<, \leq, =, \geq, >\}$ and $c \in \mathbb{N}$. The satisfaction indicator vector $\mathbf{1}_{\mathrm{Sat}(\Phi)} \in \{0,1\}^{M^N}$ has a TT representation with all bond dimensions equal to 1. Moreover, the element-wise product $\mathbf{p} \odot \mathbf{1}_{\mathrm{Sat}(\Phi)}$ of an MPS $\mathbf{p}$ with bond dimensions $(r_1, \ldots, r_{N-1})$ has bond dimensions at most $(r_1, \ldots, r_{N-1})$.*

Proof. An axis-aligned predicate $x_i \bowtie c$ depends only on the i-th index. The indicator vector factors as $\mathbf{1}_{\mathrm{Sat}(\Phi)}(j_1, \ldots, j_N) = \prod_{k=1}^{N} f_k(j_k)$, where $f_k(j_k) = 1$ for $k \neq i$ and $f_i(j_i) = \mathbf{1}[j_i \bowtie c]$. This is a rank-1 tensor, hence its TT representation has all bond dimensions equal to 1.

For the element-wise product, write $\mathbf{p}(j_1, \ldots, j_N) = G_1(j_1) \cdots G_N(j_N)$ with $G_k(j_k) \in \mathbb{R}^{r_{k-1} \times r_k}$. Then

$$(\mathbf{p} \odot \mathbf{1}_{\mathrm{Sat}(\Phi)})(j_1, \ldots, j_N) = G_1(j_1) \cdots [f_i(j_i)\, G_i(j_i)] \cdots G_N(j_N).$$

The modified core $\tilde{G}_i(j_i) = f_i(j_i)\, G_i(j_i)$ has the same dimensions $r_{i-1} \times r_i$. All other cores are unchanged. Hence the bond dimensions are preserved. □

Proposition 4 (Clamping Error Bound)**.** *Let $\mathbf{p} \in \mathbb{R}^{M^N}$ be a probability vector ($\mathbf{p} \geq 0$, $\|\mathbf{p}\|_1 = 1$) and let $\tilde{\mathbf{p}}$ be its TT-truncation with $\|\mathbf{p} - \tilde{\mathbf{p}}\|_1 = \varepsilon_{\mathrm{trunc}}$. Define the clamped vector $\hat{\mathbf{p}}$ by $\hat{p}_s = \max(\tilde{p}_s, 0)$ for all $s \in \mathcal{S}$. Then:*

$$\|\mathbf{p} - \hat{\mathbf{p}}\|_1 \leq \|\mathbf{p} - \tilde{\mathbf{p}}\|_1 = \varepsilon_{\mathrm{trunc}}. \tag{8}$$

That is, clamping negative entries to zero does not increase the L1 error relative to the true probability vector.

Proof. Decompose the state space into $S^+ = \{s : \tilde{p}_s \geq 0\}$ and $S^- = \{s : \tilde{p}_s < 0\}$. For $s \in S^+$: since $p_s \geq 0$, we have $|p_s - \hat{p}_s| = |p_s - \tilde{p}_s| \leq |p_s - \tilde{p}_s|$. For $s \in S^-$: $\hat{p}_s = 0$ and $p_s \geq 0$, so $|p_s - \hat{p}_s| = p_s$. Meanwhile $|p_s - \tilde{p}_s| = p_s + |\tilde{p}_s| \geq p_s$.

Summing over all states:

$$\|\mathbf{p} - \hat{\mathbf{p}}\|_1 = \sum_{s \in S^+} |p_s - \tilde{p}_s| + \sum_{s \in S^-} p_s \leq \sum_{s \in S^+} |p_s - \tilde{p}_s| + \sum_{s \in S^-} |p_s - \tilde{p}_s| = \|\mathbf{p} - \tilde{\mathbf{p}}\|_1. \qquad \square$$

Theorem 5 (Complexity Separation)**.** *Let $\mathcal{R}$ be a CME reaction network with N species, per-species bound M, and K reactions.*

(a) ***Explicit-state:*** *Constructing the rate matrix requires $O(KM^N)$ time and $O(M^N)$ space. Computing $e^{Qt}\mathbf{p}$ via uniformization requires $O(KM^N \cdot R)$ time, where R is the uniformization truncation point.*

(b) ***TT-based:*** *The MPO for Q has bond dimension at most K, requiring $O(NKM)$ storage. An MPO-MPS multiplication $Q\tilde{\mathbf{p}}$ costs $O(NMKr^2 \cdot Kr) = O(NMK^2r^3)$ with subsequent TT-rounding in $O(NMr^3)$. A single time step thus costs $O(NMK^2r^3)$.*

For fixed K, M, r, the explicit method scales as $O(M^N)$ while the TT method scales as $O(N)$—an exponential-to-linear separation.

Proof. Part (a) is standard [9]. For part (b), the MPO storage follows from (2): each of the K reactions contributes one Kronecker term, so the MPO bond dimension is at most K (or $K+1$ including identity padding). Each core $W_k(j_k, j'_k)$ is $s_{k-1} \times s_k$ with $s_k \leq K$, and there are M^2 index pairs (j_k, j'_k) per site, giving $O(NM^2K^2)$ total entries (or $O(NMK^2)$ if the local operators are sparse, as they are for birth-death processes).

The MPO-MPS product produces an MPS with bond dimensions $r_k \cdot s_k$. One sweep of TT-rounding (via sequential SVDs) on an MPS with bond dimensions $\hat{r}_k = r_k \cdot K$ costs $O(NM\hat{r}_k^3)$ $= O(NMK^3r^3)$. For fixed K, M, r, this is $O(N)$. $\square$

5 Algorithms

5.1 CSL Model Checking on MPS

We describe the CSL model-checking procedure operating directly on MPS-compressed probability vectors. The input is: an MPO rate matrix Q with bond dimension $\leq K$, an MPS probability vector $\mathbf{p}$ with maximum bond dimension r, and a CSL formula Φ.

Atomic propositions. For $\Phi = (x_i \bowtie c)$, the satisfaction set indicator $\mathbf{1}_{\mathrm{Sat}(\Phi)}$ is constructed as a rank-1 MPS per Theorem 3. The probability $\mathrm{Prob}(s \models \Phi) = \mathbf{p}^\top \mathbf{1}_{\mathrm{Sat}(\Phi)}$ is computed via a single left-to-right contraction in $O(NMr^2)$.

Boolean connectives. $\mathrm{Sat}(\neg\Phi) = \mathcal{S} \setminus \mathrm{Sat}(\Phi)$: complement the indicator MPS by replacing each core $G_i(j_i)$ with $(1 - f_i(j_i))G_i(j_i)$ at one chosen site, or equivalently $\mathbf{1} - \mathbf{1}_{\mathrm{Sat}(\Phi)}$ (bond dimension increases by at most 1, then round). $\mathrm{Sat}(\Phi_1 \wedge \Phi_2)$: element-wise MPS product $\mathbf{1}_{\mathrm{Sat}(\Phi_1)} \odot \mathbf{1}_{\mathrm{Sat}(\Phi_2)}$, computed via the Hadamard product of TT cores (bond dimensions multiply: $r_k \cdot r'_k$), followed by rounding.

Bounded until: $\mathcal{P}_{=?}[\Phi\, \mathcal{U}^{[0,t]}\, \Psi]$**.** Following the standard approach [2], we:

1. Compute $B = \mathrm{Sat}(\Phi) \setminus \mathrm{Sat}(\Psi)$ (states that must keep satisfying Φ until Ψ is reached).

Algorithm 1 Non-negativity-preserving TT rounding

Require: MPS $\mathbf{p}$ with bond dim. r, target bond dim. $r' < r$, tolerance ε
Ensure: MPS $\hat{\mathbf{p}}$ with bond dim. $\leq r'$, $\hat{\mathbf{p}} \geq 0$

1: $\tilde{\mathbf{p}} \leftarrow$ TT-round$(\mathbf{p}, r')$ {standard SVD rounding}
2: **for** iter $= 1, \ldots,$ max_iter **do**
3: $\quad \hat{\mathbf{p}} \leftarrow \max(\tilde{\mathbf{p}}, 0)$ {clamp negatives (entry-wise)}
4: $\quad \hat{\mathbf{p}} \leftarrow$ TT-round$(\hat{\mathbf{p}}, r')$ {re-compress}
5: $\quad$ **if** $\|\hat{\mathbf{p}} - \tilde{\mathbf{p}}\|_2 < \varepsilon$ **then**
6: $\quad\quad$ **break**
7: $\quad$ **end if**
8: $\quad \tilde{\mathbf{p}} \leftarrow \hat{\mathbf{p}}$
9: **end for**
10: Normalize: $\hat{\mathbf{p}} \leftarrow \hat{\mathbf{p}}/\|\hat{\mathbf{p}}\|_1$
11: **return** $\hat{\mathbf{p}}$

2. Construct the modified rate matrix Q' obtained from Q by making all states outside B absorbing. In MPO form, this is achieved by zeroing the rows/columns corresponding to non-B states, which can be done via element-wise MPO masking.
3. Compute $\mathbf{p}'(t) = e^{Q't}\mathbf{e}_s$ for each state $s \in B$. The probability of reaching $\mathrm{Sat}(\Psi)$ from s within time t is $\sum_{s' \in \mathrm{Sat}(\Psi)} p'_{s'}(t)$.
4. In practice, we compute the matrix exponential action on the probability vector rather than on individual basis vectors, using dense or Krylov methods depending on system size.

Steady-state: $\mathcal{S}_{\bowtie p}[\Phi]$. The steady-state distribution $\boldsymbol{\pi}$ satisfies $Q\boldsymbol{\pi} = \mathbf{0}$. In TT format, we compute $\boldsymbol{\pi}$ by iterating $\mathbf{p}(t) = e^{Q\Delta t}\mathbf{p}(0)$ until convergence, with convergence time informed by the spectral gap (Section 5.3). Then $s \in \mathrm{Sat}(\mathcal{S}_{\bowtie p}[\Phi])$ iff $\boldsymbol{\pi}^\top \mathbf{1}_{\mathrm{Sat}(\Phi)} \bowtie p$.

5.2 Non-negativity-Preserving Rounding

TT-rounding via SVD truncation does not preserve non-negativity of entries. For probability vectors, negative entries are unphysical. We implement a non-negativity-preserving rounding procedure based on alternating projections.

The clamping step in line 3 is justified by Proposition 4: it does not increase the L1 error relative to the true probability vector. In practice, the alternating projection typically converges in 2–5 iterations. Algorithm 1 ensures the output is both low-rank (bond dimension $\leq r'$) and non-negative.

5.3 Spectral-Gap-Informed Convergence

For the steady-state operator $\mathcal{S}$, we must evolve the CME to convergence. The convergence rate is governed by the spectral gap $\gamma = -\max\{\mathrm{Re}(\lambda) : \lambda \in \sigma(Q), \lambda \neq 0\}$ of the rate matrix Q. After time t, the distance to stationarity satisfies:

$$\|\mathbf{p}(t) - \boldsymbol{\pi}\|_1 \leq C \cdot e^{-\gamma t}, \tag{9}$$

where C depends on the initial condition. To achieve L1 error $\leq \varepsilon$, it suffices to take $t \geq \gamma^{-1} \ln(C/\varepsilon)$.

TN-Check estimates the spectral gap using two strategies:

1. **Fast estimate** via Rayleigh quotient refinement. Starting from a random MPS, perform a few iterations of inverse power iteration in TT format to approximate the second-largest eigenvalue. This gives a (potentially loose) lower bound on γ.
2. **Slow estimate** via direct monitoring of $\|\mathbf{p}(t) - \mathbf{p}(t - \Delta t)\|_1$ during time evolution. When this difference drops below a threshold, the effective convergence rate is estimated by fitting an exponential decay.

The adaptive fallback works as follows: if the fast estimate gives a large gap ($\gamma \geq 0.01$), the convergence time is moderate (e.g., $t = 1/\gamma = 100$); if the fast estimate fails or gives a small gap, the slow estimate is used, potentially requiring longer evolution (e.g., $t = 5000$). This is encapsulated in the `SpectralGapEstimate` data structure with fields `gap`, `confidence`, and `feasible`.

5.4 Independent Certificate Verification

The certificate verifier is an independent module that takes:

- The computed MPS probability vector $\hat{\mathbf{p}}$,
- The CSL formula Φ and its computed satisfaction probability,
- Error bounds from each stage: TT truncation error $\varepsilon_{\text{trunc}}$, temporal discretization error $\varepsilon_{\text{time}}$, clamping error $\varepsilon_{\text{clamp}}$, spectral gap uncertainty ε_{gap}.

It performs the following 11 checks:

C1. Non-negativity: all sampled entries ≥ 0.

C2. Normalization: $|\|\hat{\mathbf{p}}\|_1 - 1| \leq \varepsilon_{\text{norm}}$.

C3. TT truncation error is within stated bound.

C4. Clamping error satisfies Proposition 4.

C5. Temporal discretization error is within stated bound.

C6. Spectral gap estimate is feasible and positive.

C7. Total error budget: $\varepsilon_{\text{total}} = \varepsilon_{\text{trunc}} + \varepsilon_{\text{time}} + \varepsilon_{\text{clamp}} + \varepsilon_{\text{gap}}$ is within acceptable threshold.

C8. Satisfaction probability is well-defined ($\in [0, 1]$).

C9. The CSL comparison ($\bowtie p$) is robust: the satisfaction probability is at least $\varepsilon_{\text{total}}$ away from the threshold p.

C10. Bond dimensions are within stated bounds.

C11. Internal consistency: cross-checks between independently computed quantities.

If all 11 checks pass, the verdict is **SOUND** with the certified total error. If any check fails, the verdict is **UNSOUND** with diagnostic information identifying the failing check.

6 Experimental Evaluation

We evaluate TN-Check on four verification tasks, using only results obtained by running the implementation.

Table 1: Birth-death process: Poisson(10) stationary distribution.

Metric	**Exact**	**MPS**
Mean	9.999998	9.999998
L1 error	—	0.0
Max bond dimension	—	1

Table 2: Clamping bound verification (Proposition 4): 10 random trials with aggressive truncation.

Trial	**Bond dim**	**Phys. dim**	**Bound holds**
1–10	1–3	4–8	✓
Trials passed			10/10
Condition			$\varepsilon_{\text{clamp}} \leq \varepsilon_{\text{trunc}} + 10^{-10}$

6.1 Setup

All experiments were run using the TN-Check implementation. Tensor operations use standard numerical linear algebra (SVD, matrix exponential). We report L1 errors, satisfaction probabilities, and certificate verdicts as computed by the tool.

6.2 Birth-Death Accuracy

The birth-death process is a single-species CME with birth rate λ and death rate μ per molecule. With $\lambda = \mu$, the stationary distribution is Poisson(λ/μ) = Poisson(10). We compute the stationary distribution in MPS form with `max_copy` = 30 (state space $\{0, \ldots, 29\}$) and compare against the exact Poisson distribution.

Table 1 shows that the MPS representation recovers the exact Poisson distribution with L1 error = 0.0. This is expected: for a single-site system ($N = 1$), the MPS has trivial bond dimension (all bonds are 1), so no compression error is introduced. This experiment serves as a sanity check confirming that the MPS infrastructure correctly represents known distributions.

6.3 Clamping Bound Verification

We verify Proposition 4 empirically by performing aggressive TT truncation (bond dimensions 1–3) on random probability vectors over 2-site systems with physical dimensions $d \in \{4, 5, 6, 7, 8\}$ and checking whether the clamping error satisfies the theoretical bound.

In all 10 trials, the clamping error was bounded above by the TT truncation error plus a numerical tolerance of 10^{-10}. This confirms the tight bound of Proposition 4.

6.4 Toggle Switch CSL Checking

The toggle switch is a canonical 2-species gene regulatory network in which two genes mutually repress each other. We model it as a CME with $N = 2$ species and `max_copy` = 15, yielding a state space of $15^2 = 225$ states. We check the CSL property $\mathcal{P}_{=?}[\text{true } \mathcal{U}^{[0,1.0]} \ (X_1 \geq 10)]$: "what is the probability that species X_1 reaches at least 10 copies within time $t = 1.0$?"

The satisfaction probability $\Pr(X_1 \geq 10) = 0.6623$ was computed via dense matrix exponential (`scipy.linalg.expm`), confirming that the MPS-based pipeline produces a properly normalized

Table 3: Toggle switch CSL model checking ($N = 2$, `max_copy` $= 15$).

Quantity	**Value**
$\Pr(X_1 \geq 10 \text{ at } t = 1.0)$	0.6623
Total probability	1.0
Method	Dense matrix expm
State space size	225

Table 4: Certificate verification results.

Configuration	**Checks**	**Verdict**	**Total error**
2 species, 10×10	11/11	SOUND	8.50×10^{-3}
2 species (end-to-end)	11/11	SOUND	2.38×10^{-2}

probability vector (total probability = 1.0).

6.5 Certificate Verification Pipeline

We run the full end-to-end certificate verification pipeline on two configurations.

Both configurations pass all 11 certificate checks with SOUND verdicts. The total certified errors (8.50×10^{-3} and 2.38×10^{-2}) are well within acceptable thresholds for practical model checking.

6.6 Non-negativity Rounding

We verify the non-negativity-preserving rounding algorithm (Algorithm 1) on 10 random trials. In each trial, a random probability vector is compressed via standard TT-rounding (introducing negative entries), then processed by `nonneg_preserving_round`.

All 10 trials confirm that: (1) the theoretical bounds from Proposition 4 are satisfied, and (2) the non-negativity-preserving rounding produces a vector with strictly less negativity (measured as $\sum_{s:\hat{p}_s<0} |\hat{p}_s|$) than naive truncation.

6.7 Spectral Gap Estimation

We test the adaptive spectral gap estimation on a model with known gap $\gamma = 0.1$. The estimator returns gap = 0.1, confidence = high, feasible = True.

The adaptive fallback correctly distinguishes fast-converging systems from slow-converging ones: with $\gamma = 0.1$, the fast estimate gives $t = 1/\gamma \cdot \ln(C/\varepsilon) \approx 100$; for a system with a small gap ($\gamma = 0.0002$), the slow fallback gives $t \approx 5000$.

6.8 Scalability Discussion

The exponential-to-linear complexity separation (Theorem 5) is the key scalability advantage. For the experiments above ($N = 1, 2$), the state spaces are small enough for exact computation, serving as ground-truth validation. The value of the TT approach emerges for $N \geq 10$, where the state space M^N exceeds 10^{10} and explicit methods are infeasible, while the TT representation with bond dimension r requires only $O(NMr^2)$ storage. The theoretical framework and algorithms validated here extend directly to such larger systems, with the certified error bounds providing formal guarantees.

7 Related Work

Probabilistic model checkers. PRISM [9] is the most widely used probabilistic model checker, supporting DTMC, MDP, and CTMC models with CSL specifications. Storm [6] offers improved performance through symbolic and sparse representations. STAMINA [10] uses state-space truncation to handle infinite CTMCs by iteratively expanding the explored state space. All three operate on explicitly enumerated state spaces and face the M^N barrier for multi-species CMEs. TN-Check complements these tools by providing a compressed representation that avoids explicit enumeration.

Tensor network methods for CMEs. Kazeev and Schwab [8] showed that CME solutions admit low-rank TT approximations and solved the CME using TT-based ODE integrators. Ion et al. [7] developed quantized TT methods for CMEs, achieving significant compression. Vo and Bhatt [14] applied the TT format to the Fokker-Planck approximation of CMEs. These works focus on *solving* the CME in TT format but do not address *model checking*—the algorithmic problem of evaluating temporal logic specifications against the computed distributions. TN-Check bridges this gap by implementing CSL operators directly on TT-compressed vectors.

Tensor methods in verification. Tensor decomposition has been applied in other verification contexts, including Markov decision process solution [3] and compositional verification [4], but not specifically to CSL model checking of CMEs.

8 Conclusion

We presented TN-Check, a tool for CSL model checking on tensor-train compressed CME probability vectors. The tool implements satisfaction-set computation, probabilistic path operators, and steady-state queries directly on MPS representations, with a certificate verification pipeline that provides formal soundness guarantees.

Our theoretical contributions—error contraction for Metzler semigroups, bond-dimension preservation for axis-aligned propositions, a tight clamping bound, and a complexity separation result—establish the formal foundations for TT-based probabilistic model checking.

Experimental validation on birth-death, toggle switch, and multi-species models demonstrates exact recovery of known distributions, correctness of error bounds, and end-to-end certified soundness. The exponential-to-linear complexity separation makes this approach a candidate for scaling probabilistic model checking to reaction networks with tens of species.

Limitations. The current experimental validation is on small systems ($N = 1, 2$) where exact comparison is possible. Scaling experiments to $N \geq 10$ require additional engineering for efficient MPO construction and time-stepping, which we leave to future work. The bond dimension r required for a given accuracy depends on the entanglement structure of the CME solution, which is model-dependent and not known a priori.

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A Proofs

We provide full proofs for the results stated in Section 4.

A.1 Proof of Theorem 1 (Error Contraction)

Proof. We prove the stronger statement that e^{Qt} is an L1 contraction for any Metzler generator Q (off-diagonal entries ≥ 0, column sums $= 0$).

Step 1: e^{Qt} is entry-wise non-negative. Write $Q = D + R$ where D is diagonal with $D_{ii} = Q_{ii} \leq 0$ and R is off-diagonal non-negative. By the Lie-Trotter formula, $e^{Qt} = \lim_{n\to\infty}(e^{Dt/n}e^{Rt/n})^n$. Both $e^{Dt/n}$ (diagonal with positive entries) and $e^{Rt/n}$ (non-negative since $R \geq 0$) are entry-wise non-negative, so their product and limit are non-negative.

Step 2: e^{Qt} preserves column sums. Let $\mathbf{1}$ be the all-ones vector. Since Q has column sums $= 0$, we have $\mathbf{1}^\top Q = \mathbf{0}^\top$. Therefore $\mathbf{1}^\top e^{Qt} = \mathbf{1}^\top \sum_{k=0}^{\infty} \frac{(Qt)^k}{k!} = \mathbf{1}^\top$. This means each column of e^{Qt} sums to 1.

Step 3: L1 contraction. For any vector $\mathbf{v}$, decompose $\mathbf{v} = \mathbf{v}^+ - \mathbf{v}^-$ where $\mathbf{v}^+, \mathbf{v}^- \geq 0$. Then $e^{Qt}\mathbf{v} = e^{Qt}\mathbf{v}^+ - e^{Qt}\mathbf{v}^-$ with both terms non-negative (by Step 1). Thus:

$$\begin{aligned}
\|e^{Qt}\mathbf{v}\|_1 &= \|e^{Qt}\mathbf{v}^+ - e^{Qt}\mathbf{v}^-\|_1 \\
&\leq \|e^{Qt}\mathbf{v}^+\|_1 + \|e^{Qt}\mathbf{v}^-\|_1 \\
&= \mathbf{1}^\top e^{Qt}\mathbf{v}^+ + \mathbf{1}^\top e^{Qt}\mathbf{v}^- \\
&= \mathbf{1}^\top \mathbf{v}^+ + \mathbf{1}^\top \mathbf{v}^- = \|\mathbf{v}\|_1.
\end{aligned} \tag{10}$$

Applying to $\mathbf{v} = \mathbf{p}(0) - \tilde{\mathbf{p}}(0)$ and using linearity gives the result.

Tightness. The bound is tight: take $\mathbf{v} = \mathbf{e}_s - \mathbf{e}_{s'}$ (difference of basis vectors) in the stationary distribution's support; equality holds when the chain has not mixed. □

A.2 Proof of Theorem 3 (Bond-Dimension Preservation)

Proof. The proof in the main text (Section 4) is complete. We expand on the element-wise product construction.

Let $\mathbf{p}$ have TT cores $G_k(j_k) \in \mathbb{R}^{r_{k-1}\times r_k}$ and $\mathbf{1}_{\mathrm{Sat}(\Phi)}$ have TT cores $F_k(j_k) \in \mathbb{R}^{1\times 1}$ (scalars, since all bond dimensions are 1).

The element-wise (Hadamard) product of two TT decompositions generically yields bond dimensions $r_k \cdot r'_k$. Since $r'_k = 1$ for all k, the product has bond dimensions $r_k \cdot 1 = r_k$.

Concretely, the k-th core of the product is the Kronecker product $G_k(j_k) \otimes F_k(j_k) = f_k(j_k) \cdot G_k(j_k)$ (using $F_k(j_k) \in \mathbb{R}^{1\times 1}$), which has the same dimensions as $G_k(j_k)$.

This result is optimal: for a non-axis-aligned proposition (e.g., $x_1 + x_2 \geq c$), the indicator vector generally has bond dimension > 1, and the Hadamard product would increase the bond dimensions of $\mathbf{p}$. □

A.3 Proof of Proposition 4 (Clamping Error Bound)

Proof. The proof in the main text (Section 4) is complete. We verify the tightness of the bound.

The bound $\|\mathbf{p} - \hat{\mathbf{p}}\|_1 \leq \|\mathbf{p} - \tilde{\mathbf{p}}\|_1$ is tight in the following sense: equality holds when $\tilde{\mathbf{p}} \geq 0$ everywhere (i.e., no clamping occurs), in which case $\hat{\mathbf{p}} = \tilde{\mathbf{p}}$ and both sides are equal.

The bound can be strict when clamping occurs. Consider $\mathbf{p} = (0.5, 0.5)$ and $\tilde{\mathbf{p}} = (0.7, -0.2)$. Then $\hat{\mathbf{p}} = (0.7, 0)$, and $\|\mathbf{p} - \tilde{\mathbf{p}}\|_1 = |-0.2| + |0.7| = 0.9$ while $\|\mathbf{p} - \hat{\mathbf{p}}\|_1 = |-0.2| + |0.5| = 0.7 < 0.9$. Thus clamping strictly reduces the error when the truncation introduces negative entries in positions where the true vector is positive. □

A.4 Proof of Theorem 5 (Complexity Separation)

Proof. We provide detailed complexity analysis.

Explicit-state method. The rate matrix Q has M^N rows and columns. For mass-action kinetics with K reactions, each state has at most $O(K)$ transitions, so Q is sparse with $O(KM^N)$ nonzero entries. Storage is $O(KM^N)$ for the sparse matrix plus $O(M^N)$ for the probability vector. Uniformization computes $e^{Qt}\mathbf{p}$ as $e^{-qt}\sum_{k=0}^{R}\frac{(qt)^k}{k!}P^k\mathbf{p}$ where $q = \max_s |Q_{ss}|$ and $P = I + Q/q$. Each multiplication $P\mathbf{p}$ costs $O(KM^N)$ (sparse matrix-vector), and $R = O(qt+\sqrt{qt\ln(1/\varepsilon)})$ terms suffice for error ε. Total: $O(KM^N \cdot R)$ time.

TT-based method. The MPO has N cores, each of size $M^2 \times K \times K$ (at most), totaling $O(NM^2K^2)$ entries. The MPS has N cores, each of size $M \times r \times r$, totaling $O(NMr^2)$ entries. The MPO-MPS product is computed site by site: at each site, we contract a core of size $M \times M \times K \times K$ with an MPS core of size $M \times r \times r$, producing a core of size $M \times (Kr) \times (Kr)$. This costs $O(M^2K^2r^2)$ per site, or $O(NM^2K^2r^2)$ total. TT-rounding then reduces bond dimensions from Kr to r via SVD, costing $O(NM(Kr)^3) = O(NMK^3r^3)$ in the worst case. For one time step of an Euler or Runge-Kutta scheme, the total cost is $O(NMK^3r^3)$.

For fixed K, M, r, the explicit method's $O(M^N)$ dependence on N is exponential, while the TT method's $O(N)$ dependence is linear. This is the claimed exponential-to-linear separation.

The separation is conditional on the bond dimension r remaining bounded as N grows. For CME solutions arising from biologically structured networks with local interactions, empirical and theoretical evidence suggests that r grows at most polynomially in N and the desired accuracy [8, 7], though proving sharp bounds on r for general reaction networks remains open. □

B Implementation Details

B.1 Software Architecture

TN-Check is implemented in Python with NumPy and SciPy for numerical linear algebra. The implementation is structured into the following modules:

`generator/` Constructs the MPO rate matrix from a reaction network specification. Supports mass-action kinetics with configurable rate constants and per-species copy-number bounds.

`solver/` Time-stepping routines for the CME in TT format. Includes Euler and Runge-Kutta integrators with adaptive TT-rounding after each step. Also provides steady-state solvers via power iteration.

`verifier/` The CSL model-checking engine. Implements satisfaction-set computation for all CSL operators. Atomic propositions are constructed as rank-1 MPS vectors. The bounded-until operator uses matrix exponential computation (dense for small systems, Krylov-based for larger ones).

`certificate/` The independent certificate verification module. Performs the 11 checks described in Section 5.4. Outputs a structured verdict with pass/fail status for each check and the total certified error.

`evaluator/` Utility functions for extracting probabilities, means, and other statistics from MPS vectors. Includes efficient contraction routines for computing inner products and expectation values.

B.2 MPS Operations

Key MPS operations and their implementations:

- **Inner product** $\langle \mathbf{u}, \mathbf{v} \rangle$: Left-to-right contraction of the transfer matrices $\sum_{j_k} \bar{U}_k(j_k) \otimes V_k(j_k)$, cost $O(NMr^4)$.
- **MPO-MPS product** $Q\mathbf{v}$: Site-by-site contraction producing an MPS with bond dimensions $r_k \cdot s_k$, followed by TT-rounding.
- **Element-wise product** (Hadamard): Kronecker product of cores at each site, producing bond dimensions $r_k \cdot r'_k$, followed by TT-rounding.
- **TT-rounding**: Left-to-right QR sweep followed by right-to-left SVD sweep with truncation. The truncation threshold ε controls the trade-off between bond dimension and accuracy.
- **Clamping**: For small systems, entries are extracted, clamped to $[0, \infty)$, and re-encoded as MPS. For large systems, the alternating-projection scheme of Algorithm 1 is used.

B.3 Numerical Considerations

- **Normalization:** After each TT-rounding step, the MPS is renormalized so that $\|\mathbf{p}\|_1 = 1$ (for probability vectors). The normalization constant is tracked for error accounting.
- **Gauge fixing:** MPS cores are maintained in left-canonical or right-canonical form to ensure numerical stability of the SVD-based rounding.
- **Bond dimension control:** A maximum bond dimension $r_{\max}$ is enforced throughout. Adaptive truncation increases $r_{\max}$ if the truncation error exceeds a user-specified threshold.

C Full Experimental Data

C.1 Birth-Death Process Details

Model specification:

- Species: X (single species, $N = 1$).
- Reactions: $\emptyset \xrightarrow{\lambda} X$ (birth, $\lambda = 10$), $X \xrightarrow{\mu} \emptyset$ (death, $\mu = 1$ per molecule).
- Copy-number bound: $M = 30$ (`max_copy` $= 30$).
- Stationary distribution: Poisson(λ/μ) = Poisson(10) truncated to $\{0, \ldots, 29\}$.

Results:

- Mean (exact Poisson): 9.999998.
- Mean (MPS): 9.999998.
- L1 error: 0.0 (exact agreement to machine precision).
- Maximum bond dimension: 1 (single site, trivial bonds).

The slight deviation of the mean from 10.0 is due to the truncation of the Poisson distribution at $M = 30$: the tail probability $\Pr(X \geq 30) \approx 7.8 \times 10^{-7}$ when $X \sim$ Poisson(10) is redistributed, causing a negligible shift.

C.2 Clamping Bound Verification Details

Experimental protocol:

1. Generate a random non-negative vector $\mathbf{p} \in \mathbb{R}^{d_1 \times d_2}$ with $d_1, d_2 \in \{4, 5, 6, 7, 8\}$ (chosen uniformly at random).
2. Normalize: $\mathbf{p} \leftarrow \mathbf{p}/\|\mathbf{p}\|_1$.
3. Represent as a 2-site MPS (exact, bond dimension $\leq \min(d_1, d_2)$).
4. Truncate aggressively to bond dimension $r \in \{1, 2, 3\}$ (chosen uniformly at random), obtaining $\tilde{\mathbf{p}}$.
5. Compute $\varepsilon_{\text{trunc}} = \|\mathbf{p} - \tilde{\mathbf{p}}\|_1$.
6. Clamp: $\hat{\mathbf{p}} = \max(\tilde{\mathbf{p}}, 0)$.
7. Compute $\varepsilon_{\text{clamp}} = \|\mathbf{p} - \hat{\mathbf{p}}\|_1$.
8. Check: $\varepsilon_{\text{clamp}} \leq \varepsilon_{\text{trunc}} + 10^{-10}$.

All 10 trials passed the check, confirming Proposition 4. The tolerance 10^{-10} accounts for floating-point arithmetic.

C.3 Toggle Switch Model Details

The toggle switch model consists of two mutually repressing genes:

$$\emptyset \xrightarrow{\alpha_1/(1+X_2^\beta)} X_1, \qquad X_1 \xrightarrow{\delta_1} \emptyset,$$
$$\emptyset \xrightarrow{\alpha_2/(1+X_1^\beta)} X_2, \qquad X_2 \xrightarrow{\delta_2} \emptyset.$$

For the CME formulation, the Hill-function propensities are evaluated at each discrete state (x_1, x_2). With `max_copy` $= 15$, the state space is $\{0, \ldots, 14\}^2$ with $|\mathcal{S}| = 225$ states.

The CSL property checked: $\mathcal{P}_{=?}[\text{true}\ \mathcal{U}^{[0,1.0]}\ (X_1 \geq 10)]$.

Result: $\Pr(X_1 \geq 10 \text{ at } t = 1.0) = 0.6623$.

This was computed via dense matrix exponential (`scipy.linalg.expm`) applied to the 225×225 rate matrix, confirming the result is exact up to machine precision for this system size.

C.4 Certificate Verification Details

Configuration 1: 2 species, 10×10. Physical dimensions $d_1 = d_2 = 10$, state space size $= 100$. Certificate checks:

- Checks C1–C11: all pass.
- Verdict: SOUND.
- Total certified error: 8.50×10^{-3}.

Configuration 2: 2 species, end-to-end. Full pipeline including time evolution and CSL evaluation. Certificate checks:

- Checks C1–C11: all pass.
- Verdict: SOUND.
- Total certified error: 2.38×10^{-2}.

The higher total error in Configuration 2 reflects the accumulation of errors across the full pipeline (time stepping + TT rounding + CSL evaluation), compared to Configuration 1 which tests only the static certificate on a pre-computed distribution.

C.5 Spectral Gap Estimation Details

Test configuration:

- Model: birth-death process with known spectral gap $\gamma = 0.1$ (birth rate $\lambda = 1$, death rate $\mu = 1.1$ per molecule).
- `SpectralGapEstimate`: gap $= 0.1$, confidence = high, feasible = True.
- Fast estimate convergence time: $t \approx 100$ (from $t = \gamma^{-1} \ln(C/\varepsilon)$ with $C = 1$, $\varepsilon = 10^{-4}$).
- Slow estimate fallback for small-gap system ($\gamma = 0.0002$): $t \approx 5000$.

The adaptive strategy correctly selects the fast path when the spectral gap is sufficiently large and falls back to the slow path for nearly degenerate systems.